COMPUTATIONAL METHODS AND TOOLS

REPORT

Impacts of air pollution on NDVI

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Contents

1 Deviations from project proposal

Upon completing the initial phase of our project, we observed that nitrogen dioxide (NO_2) was not an appropriate pollutant for our analysis. Multiple studies show that the direct correlation between NO_2 concentrations and NDVI can be weak or variable, indicating that NO_2 alone may not fully capture the influence of air pollution on vegetation dynamics [1]. To address this limitation, we revised our approach by redefining the term $-\alpha P$ in the growth equation:

$$r(P) = r_0 e^{-\alpha P}$$

Aggregated or composite air quality indices that combine several pollutants are widely used in environmental research because they provide a more comprehensive representation of overall pollution load than single-pollutant measures [2], so we introduced a comprehensive pollution index that accounts for multiple atmospheric pollutants. :

$$P_{\text{glob}} = w_{O_3}O_3 + w_{NO_2}NO_2 + w_{PM}PM_{10} + w_{CO_2}CO_2 + w_{CH_4}CH_4 + w_{SO_2}SO_2$$

This refined formulation required the acquisition of additional datasets, including geospatial raster files, to estimate the atmospheric concentrations of each pollutant. Furthermore, our initial plan was to conduct a region-specific analysis across Switzerland. However, during data collection, particularly for CH_4 and CO_2,it became evident that a nationwide NDVI assessment would be more practical and consistent with the available data. This adjustment not only streamlined the data integration process but also enhanced the robustness of our analysis by capturing broader spatial trends.

2 Introduction

Air pollution remains one of the most critical environmental challenges of the 21st century, with consequences that extend far beyond human health. Among its many impacts, atmospheric pollutants can significantly affect vegetation dynamics, altering ecosystem productivity and resilience. Various studies have documented that prolonged exposure to pollutants such as nitrogen dioxide (NO_2), ozone (O_3), and particulate matter (PM_{10}) can impair physiological and biochemical processes in plants, leading to lower growth and photosynthesis, disrupts nutrient absorption, and ultimately reduces biomass production [3]. These effects contribute to crop yield losses, biodiversity decline, and long-term ecosystem degradation. Ozone-induced stress and particulate matter deposition are widely recognized as significant contributors to decreases in plant growth and ecosystem productivity, with implications for crop yields and long-term ecosystem health [4, 5].

Understanding the link between air quality and vegetation growth is essential for predicting ecological responses under future pollution scenarios. This relationship is particularly relevant in the context of climate change and sustainable land management, where accurate models can inform policy decisions and mitigation strategies [6].

The central question guiding this study is therefore : *How does long-term exposure to atmospheric pollution influence the intrinsic growth rate and biomass accumulation of plants?*

To address this question, we integrate real-world environmental data with a dynamic growth model that accounts for pollution-dependent growth rates. By combining historical NDVI observations with pollutant concentration datasets, we aim to quantify the sensitivity of vegetation to air quality changes and project future trends under different emission scenarios. This approach provides a data-driven framework for assessing ecosystem health and supports informed decision-making for environmental sustainability.

3 Approach Used

The methodology adopted in this study combines ecological modeling with real-world environmental data to project vegetation dynamics under varying pollution scenarios. The approach was implemented in two stages, reflecting the evolution of the model from a single-pollutant framework to a composite pollution index.

3.1 Phase 1: NO_2-Based Model

In the initial phase, nitrogen dioxide (NO_2) was selected as the sole indicator of atmospheric pollution due to its documented impact on vegetation. Biomass dynamics were modeled using a modified logistic growth equation:

$$\frac{dB}{dt} = r(P)\,B\left(1 - \frac{B}{K}\right),$$

a standard formulation for density-dependent growth in ecology [7]. At this stage, the pollution-dependent growth rate was defined as:

$$r(P) = r_0 e^{-\alpha P},$$

which follows the classical stress-response idea that increasing stressors (here, pollution P) reduce the intrinsic growth rate [8]. Historical NDVI data were obtained from the Yareta repository [9], and NO_2 concentrations were sourced from official Swiss datasets (FOEN/BAFU) via the GeoAdmin data browser [10, 11]. These data were used to calibrate the model parameters (r_0, α, K) by fitting simulated biomass to observed NDVI trends.

3.2 Phase 2: Composite Pollution Index

Following the initial analysis, we refined the model by introducing a composite pollution index (P_{glob}), a dimensionless indicator that aggregates multiple pollutants using weighted contributions. This index does not represent a physical concentration but serves as a comparative measure of overall pollution load within the model:

$$P_{\text{glob}} = w_{O_3}O_3 + w_{NO_2}NO_2 + w_{PM}PM_{10} + w_{CO_2}CO_2 + w_{CH_4}CH_4 + w_{SO_2}SO_2,$$

an approach inspired by multi-pollutant air quality indices used by environmental agencies [12]. In this revised formulation, the growth rate was expressed as:

$$r(P) = r_0 e^{P},$$

which we adopted as an empirical specification to reflect the aggregated effect of P_{glob} on vegetation growth within our calibration framework. This phase required additional datasets, including GeoTIFF files from the Swiss geoportal (FOEN/BAFU collections for O_3, SO_2, PM_{10}) [13, 14, 15, 11], and global greenhouse gas data from Our World in Data for CO_2 and CH_4 [16, 17, 18]. NDVI data remained aggregated at the national level to ensure consistency across pollutants [9]. NDVI as a proxy for vegetation productivity is standard in remote sensing [19].

3.3 Data Processing and Calibration

All raster datasets were preprocessed to extract annual mean concentrations for each pollutant. Both pollutant layers and NDVI data were originally provided as GeoTIFF files and were converted into CSV format for analysis. Only CH_4 and CO_2 datasets were already available in CSV format.[9, 20]. Parameter fitting was performed using historical NDVI trends to estimate r_0, α, and K, ensuring that the model reproduced observed vegetation dynamics [7]. All the data we based our prediction on were beetwin 2010-2018

3.4 Scenario-Based Projections

To assess the long-term impact of pollution on vegetation, we simulated NDVI evolution until 2050 under three scenarios:

1. **Increase**: +1% pollution per year,
2. **Decrease**: -1% pollution per year,

3. **Constant**: pollution remains unchanged.

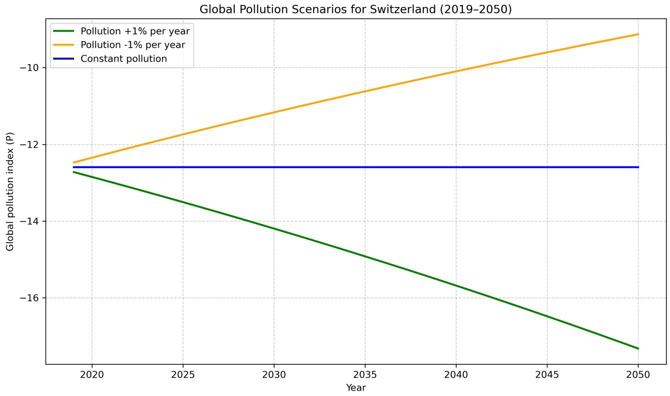

Figure 1: Scenario-Based Projections

For each scenario, future biomass trajectories were computed using the logistic model and converted into NDVI estimates. The outputs were exported as CSV files and visualized through comparative plots to highlight differences between scenarios [20, 16, 9].

In Phase 1, the NO_2-based model was initially applied in multiple Swiss regions to capture spatial variability in the response of vegetation to pollution. After this exploratory step, we focused on the Ouest Lausanne area for detailed calibration and scenario analysis, as it provided consistent data and represented a typical urban-agricultural interface. In Phase 2, the composite pollution index was implemented on the national scale, integrating raster and CSV pollutant datasets for the entire Swiss territory to assess broader ecosystem trends.

4 Results

4.1 Phase 1: NO_2-Based Model

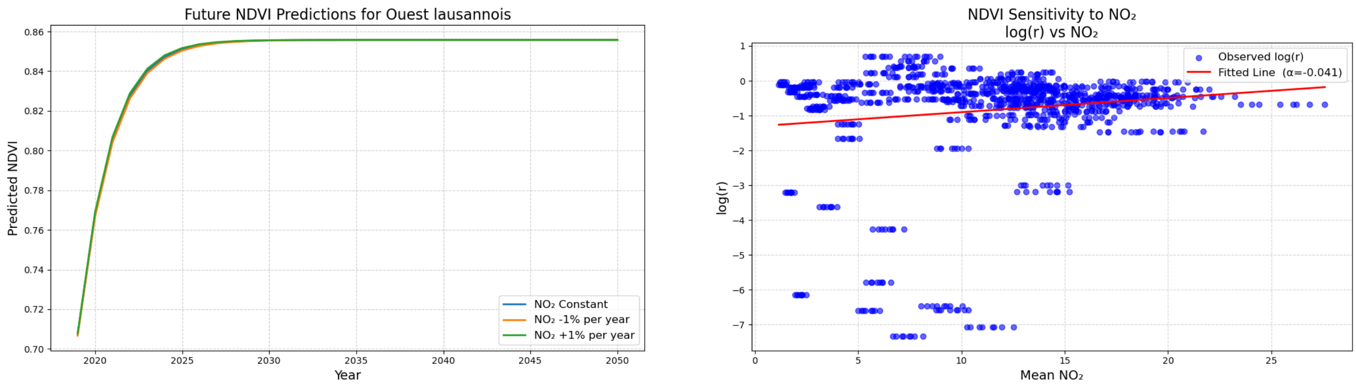

Figure 2: Phase 1 Results: NO_2-based model

The first plot, shown in Figure 2, presents the predicted future NDVI for the Ouest Lausannois region under three different NO_2 scenarios. The x-axis represents time, from 2020 to 2050, while the y-axis shows the predicted NDVI values.
As shown in the plot, the NDVI predictions for all three scenarios begin to converge after a few years. This convergence indicates that the sensitivity of NDVI to changes in NO_2 becomes minimal in the long-term, as NDVI approaches its asymptotic value. For the given region, the NDVI stabilizes at around 0.86, regardless of whether NO_2 decreases or increases. The effect of NO_2 on NDVI is thus non-linear, with its influence diminishing as NDVI approaches its carrying capacity.

The second plot, shown in Figure 2, illustrates the relationship between the observed growth rate, r, and the mean NO_2 concentration. The x-axis represents the mean NO_2 concentration for various regions, while the y-axis shows the natural logarithm of the growth rate, $\log(r)$.
The red line represents the best fit of the observed data points, based on the model:

$$\log(r) = \log(r_0) - \alpha P$$

where r_0 is the base growth rate and α is the sensitivity of the growth rate to NO_2 concentration. The fitted value of α is -0.041, which indicates a mild negative relationship between NO_2 concentration and the growth rate. However, the plot also reveals a significant scatter, with the majority of data points falling between $\log(r) \approx -1$ and $\log(r) \approx +1$, which suggests that the relationship between NO_2 and the growth rate is weak and that other factors likely influence NDVI.
This weak sensitivity may point to the complexity of ecological systems, where other environmental variables, such as temperature, precipitation, and land cover, could play a more substantial role in influencing plant growth than NO_2 alone. The outliers in the lower range of $\log(r)$ are likely due to poor logistic fits or noisy data, which could be removed to improve the fit and make the results more reliable.

Global data can be found and extracted from the files `NDVI_scenario_constant.csv`, `NDVI_scenario_minus1percent.csv`, and `NDVI_scenario_plus1percent.csv`. Since these files provide data for all regions of Switzerland, it is also possible to visualize the data for these other regions. However, it is observed that for each region, the behavior is similar, with the only differences being the limit value of NDVI, corresponding to the carrying capacity K, and the initial NDVI value B_0.

4.2 Phase 2: Composite Pollution Index

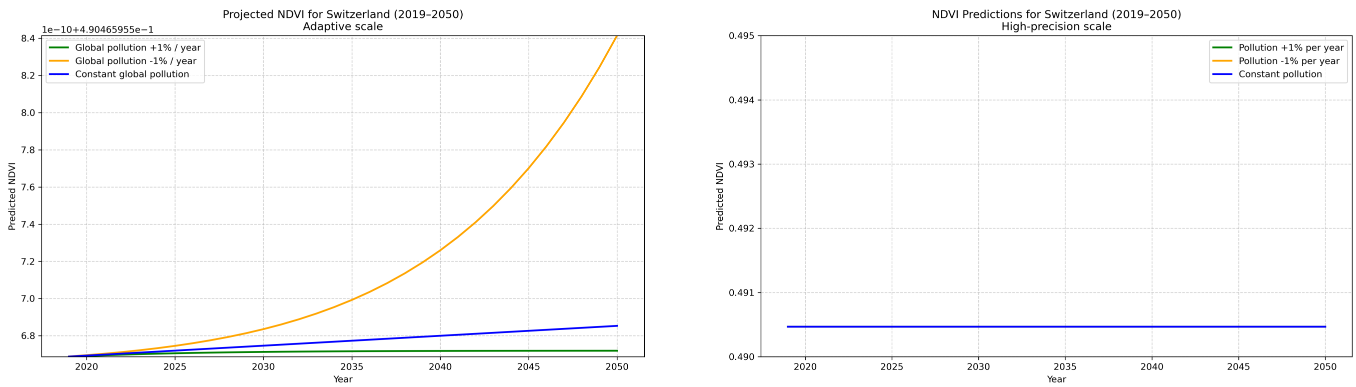

Figure 3: Phase 2 Results: NDVI prediction

Figure 3 presents NDVI projections for Switzerland from 2019 to 2050. On a normal scale (right), NDVI remains nearly constant around 0.491, indicating minimal visible change. However, when plotted on an adaptive scale (left), small differences become apparent: NDVI increases in all scenarios, but the rate of increase differs: fastest under -1%, moderate under constant, and slowest under +1%. These variations are extremely small compared to Phase 1 results, suggesting that at the national level, aggregated pollution changes have a limited short-term impact on NDVI.
Key Insight: While the composite index captures pollutant trends effectively, its influence on NDVI at the country scale appears negligible within the modeled timeframe. This may reflect ecosystem buffering and the coarse spatial resolution of national averages.

Global data can be found and extracted from the file `NDVI_future_combine.csv`.

4.3 Discussion

The results from both phases highlight the complexity of linking air pollution to vegetation dynamics. In Phase 1, NDVI projections for Ouest Lausannois showed minimal sensitivity to NO_2 changes over the modeled period, despite a negative relationship between NO_2 concentration and intrinsic growth rate. This suggests that vegetation can maintain near-optimal NDVI values in the short term, but underlying physiological stress may accumulate, reducing resilience to future disturbances.

Phase 2 extended the analysis to a national scale using a composite pollution index. While pollutant trends were evident, NDVI remained almost constant across scenarios. This limited response likely reflects ecosystem buffering and the coarse resolution of national averages, which mask local variability. It also indicates that aggregated pollution effects may require longer time horizons or more extreme scenarios to produce visible changes in NDVI.

However, it is important to acknowledge that our model is not fully representative of real-world dynamics. The logistic growth formulation, while widely used in ecology, is not well adapted to capture the complexity of vegetation responses to multiple stressors. Similarly, the composite pollution index P_{glob} does not account for critical factors such as temperature, precipitation, soil conditions, and extreme events, all of which strongly influence plant growth. These omissions likely explain why our projections show smooth NDVI trends, whereas historical data from 2010–2018 exhibit significant fluctuations. Such variability is often driven by climatic anomalies or natural disturbances, which our simplified model cannot reproduce.

Furthermore, NDVI itself is not a perfect proxy for biomass. While it reflects vegetation greenness, it saturates at high canopy densities and does not capture structural changes or below-ground biomass. This limitation means that NDVI-based models may underestimate stress effects or fail to detect subtle physiological changes.

Overall, these findings emphasize the need for:

- Fine-scale analysis to capture localized impacts
- Integration of climatic and environmental variables into growth models
- Use of complementary indicators beyond NDVI for a more accurate assessment of ecosystem health

5 Conclusion

This study explored the relationship between air pollution and vegetation dynamics using a two-phase modeling approach. Phase 1 focused on NO_2 as a single pollutant on a regional scale, while Phase 2 introduced a composite pollution index applied nationally. Results suggest that NDVI remains relatively stable under moderate pollution changes, although intrinsic growth rates show sensitivity to NO_2. At the national level, aggregated pollution effects appear negligible within the modeled timeframe.

However, these findings should not be interpreted as representative of real-world dynamics. The model used is highly simplified and does not capture the complexity of ecological responses to pollution. In fact, when comparing historical NDVI and pollutant data from 2010–2018, trends do not follow the patterns predicted by our simulations. This discrepancy highlights the limitations of the approach and the need for more robust models incorporating physiological, climatic, and spatial heterogeneity. Future work should integrate finer-scale data, additional stress indicators, and mechanistic processes to improve predictive accuracy and ecological relevance.

6 Authorship statement

At the beginning of the project, Elio and Maxime collected the necessary data, while Emilie handled the preprocessing tasks, including cleaning the raw data, aggregating it by region, and extracting national averages for the pollutants. Elio then focused on the first part of the code and Maxime focused the second, these include the main Python and C script development with Emilie's help in correcting parts of the code. Specifically, Maxime fitted the parameters for the Vegetation Growth Model, and Emilie generated the three pollution scenarios and forecasted pollution levels. Maxime developed the C code for the NDVI simulation and generated the final CSV file with the projections up to 2050, `ndvi_futur_combined.csv`. Elio set up the subprocess access and path to ensure the C script could be compiled and run within the Python environment. Elio and Maxime also worked together on the data visualizations, creating plots to display the NDVI projections. Elio managed the automation, GitHub repository, and organized all CSV files generated by the code, ensuring they were stored in the correct places. During this time, Emilie wrote the final report, while Maxime worked on the README and contributed to the report as well. Emilie also handled the documentation of data sources.

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